

AUTOMOBILE DASHBOARD ANALYSIS

Using power BI

A BUSINESS ANALYTICS PROJECT REPORT

This project analyzes automobile sales data using Power BI to identify sales trends, profitability, customer preferences, and regional performance. The dashboard transforms raw data into meaningful insights for better business decision-making

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Tool Used: Microsoft Power BI

Course: Business Analyst

Code: CX-134

INTRODUCTION

The project titled Automobile Dashboard Analysis using Power BI is designed to analyze automobile sales data and transform raw information into meaningful business insights. In today's competitive business environment, companies rely on accurate data analysis to improve performance, increase profitability, and make better strategic decisions. The automobile industry is one of the largest and most dynamic industries, where understanding customer demand, sales trends, and market performance is extremely important. This project helps in studying these factors through an interactive dashboard.

The dataset used in this project contains valuable information such as invoice dates, car brands, models, sale prices, cost prices, discounts, countries, customer names, delivery charges, spare parts cost, labor cost, and vehicle types. By using this information, different aspects of business performance can be analyzed in a structured manner. The data allows the comparison of sales across multiple brands, locations, and time periods.

Microsoft Power BI has been used as the main analytical tool for this project because it provides strong features for data cleaning, modeling, calculations, and dashboard creation. Power Query is used for transforming and preparing the dataset, while DAX formulas are used to calculate key metrics such as total sales, profit, profit margin, cumulative sales, and growth percentage. Various visuals such as bar charts, line charts, maps, pie charts, KPI cards, and matrices are used to present the findings effectively.

The main objective of this project is to identify sales patterns, top-performing brands, profitable markets, and customer preferences. It also helps understand how discounts and costs affect profitability. Through this dashboard, users can quickly access useful information and make informed decisions.

Overall, the Automobile Dashboard Analysis project demonstrates how business intelligence tools can convert complex raw data into a clear and interactive reporting system. It provides practical knowledge of data analytics and shows the importance of visualization in modern business management.

METHODOLOGY

The Automobile Dashboard Analysis project was completed using a systematic and structured methodology to ensure accurate results and meaningful insights. Microsoft Power BI Desktop was used as the primary tool for data analysis, visualization, and reporting. The methodology consisted of multiple stages including data collection, data preparation, data modeling, measure creation, dashboard development, and interpretation of results.

The first stage involved importing the automobile sales dataset into Power BI. After loading the data, the dataset was carefully reviewed to understand the available columns, data types, and overall structure. Important fields such as invoice date, brand name, model, sale price, cost price, country, and customer details were identified for analysis.

The second stage focused on data cleaning and transformation using Power Query. In this stage, unnecessary columns were removed, missing values were checked, data types were corrected, and duplicate records were eliminated. New calculated columns such as Total Cost, Profit, Profit Category, Year, and Month were created to simplify future analysis. This stage was important because clean and accurate data leads to reliable dashboard results.

The third stage involved data modeling and creating DAX measures. Measures such as Total Sales, Total Profit, Profit Margin, Average Sales, Running Total Sales, Year-to-Date Sales, Sales Growth Percentage, and Moving Average were created. These calculations helped in measuring performance across time periods, products, and regions.

The fourth stage was dashboard development. Different visuals such as KPI cards, bar charts, line charts, pie charts, maps, treemaps, scatter plots, funnels, and matrices were used to display data clearly. Slicers were also added to filter information by year, country, and brand, making the dashboard interactive and user-friendly.

The final stage included interpretation of the dashboard results and identifying business insights. Trends in sales, profitable brands, customer behavior, and discount impact were studied. The dashboard was then formatted professionally for final presentation.

REQUIREMENT ANALYSIS

Requirement analysis is an important stage of any project because it defines the objectives, expected outputs, tools required, and functional needs of the system. In the Automobile Dashboard Analysis project, the main requirement was to analyze automobile sales data and convert it into a visual dashboard that supports business decision-making. The project needed to provide clear insights regarding sales performance, profitability, customer behavior, and regional market trends.

The first requirement was the availability of a structured dataset containing automobile sales information. The provided dataset included important columns such as invoice date, sale price, cost price, discount amount, country name, customer details, car make, model, and vehicle type. These fields were necessary to perform both financial and operational analysis.

The second requirement was the use of Microsoft Power BI Desktop as the business intelligence tool. Power BI was required because it offers advanced capabilities such as Power Query for data transformation, DAX for calculations, and interactive visualizations for dashboard creation. The project also required the use of multiple chart types including bar charts, line charts, maps, pie charts, KPI cards, and matrix tables.

Another important requirement was data cleaning and preparation. Since raw data may contain inconsistencies or unnecessary fields, Power Query transformations were necessary. Tasks such as removing duplicates, changing data types, creating calculated columns, and grouping data were part of the project requirements.

The project also required advanced calculations such as total sales, total profit, profit margin, cumulative sales, moving average, sales growth percentage, and top product rankings. These measures were essential for evaluating business performance accurately.

User interaction was another requirement. The dashboard needed slicers and filters for year, country, and brand so that users could easily analyze specific data segments. Professional formatting, easy navigation, and readability were also required.

Finally, the project required proper documentation and presentation of results. Screenshots of visuals, explanation of methodology, insights, and conclusions had to be included. Overall, the requirement analysis ensured that both technical and business objectives were successfully achieved.

OTHER PARAMETERS

Other important parameters considered in this Automobile Dashboard Analysis project include data accuracy, consistency, usability, interactivity, performance, and visual presentation. Data accuracy was maintained by checking missing values, correcting data types, and validating important columns such as invoice date, sale price, cost price, discount, and country name. Consistency was ensured by using standard naming conventions for columns, uniform number formats, and common color themes throughout the dashboard. This helps users understand the report easily without confusion.

Dashboard usability was another major factor. The visuals were arranged in a structured layout so that users can quickly identify key information such as total sales, total profit, and profit margin. KPI cards were placed at the top for instant visibility, while detailed charts were organized below them. Slicers were added for year, country, and brand to make the report interactive and allow users to filter data according to their needs.

Performance optimization was also considered while building the dashboard. Necessary columns were retained, unnecessary columns were removed, and calculations were created using efficient DAX measures. This improves loading speed and smooth navigation.

Visual appeal was maintained through proper alignment, readable fonts, suitable chart titles, and professional formatting. Charts were selected based on the type of analysis required, such as line charts for trends, bar charts for comparison, maps for regional analysis, and pie charts for distribution. These parameters together improve the effectiveness, clarity, and professional quality of the final dashboard project.

VISUALIZATIONS

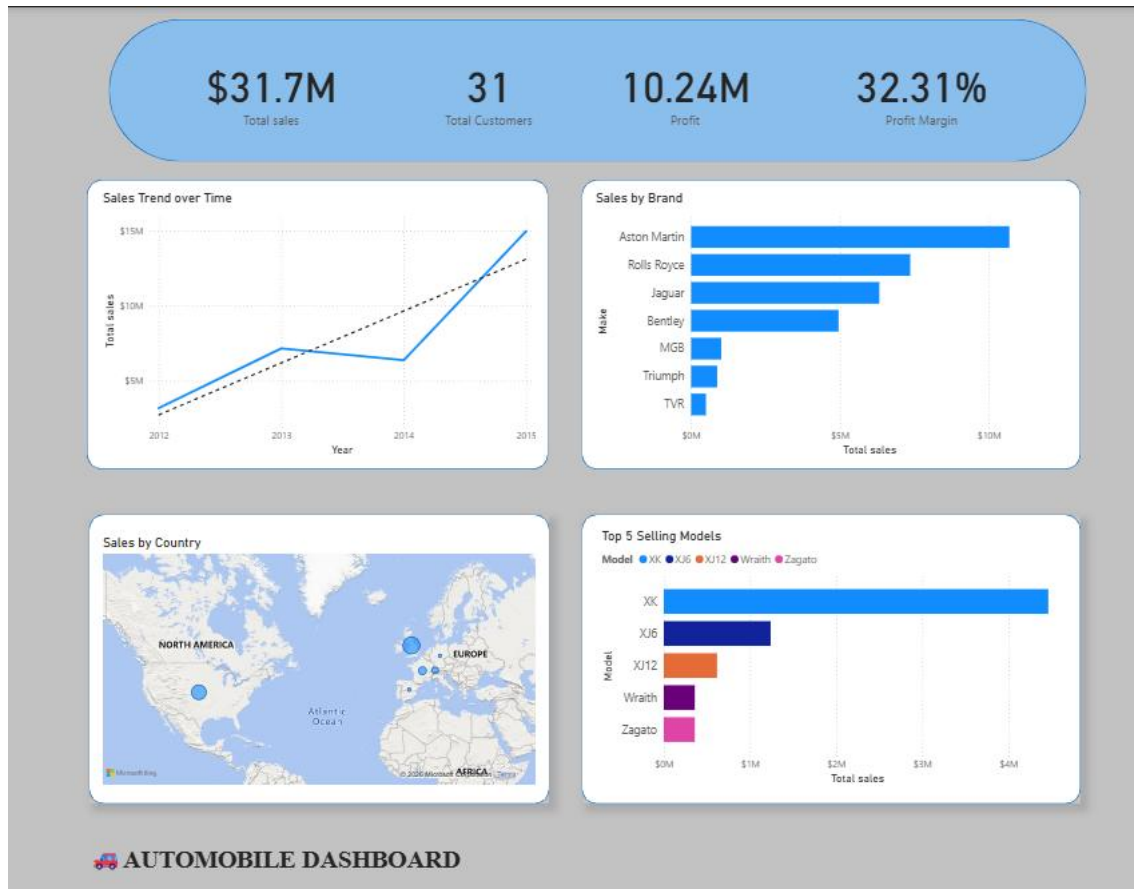


Figure 1: Dashboard Overview

This Dashboard provides a complete overview of automobile sale, profitability, customers, and regional performance by combining sales, profit, customer, product, and country-level analysis in one interactive report. It helps decision-making monitor important KPIs, identify trends, and compare performance across different dimensions efficiently.

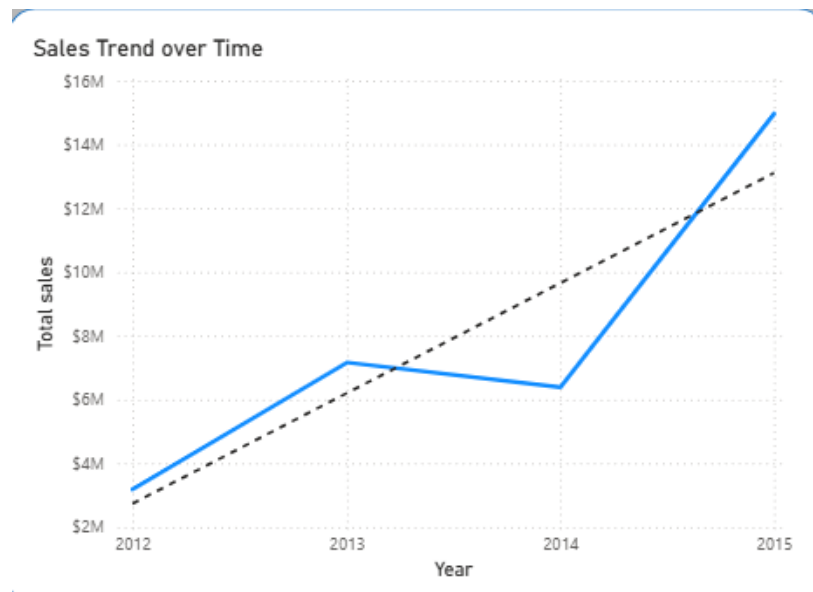


Figure 2: Sales Trend Over Time

This line chart presents the movement of total sales across different months and years. It helps in understanding seasonal demand patterns, growth periods, and low-performing months. Such trend analysis is useful for forecasting future sales and planning inventory accordingly.

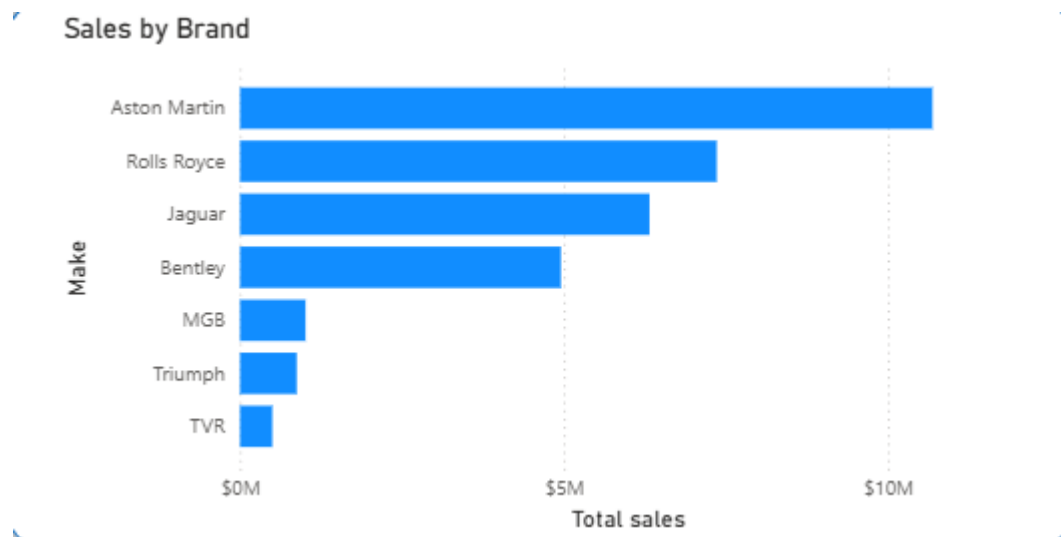


Figure 3: Sales by Brand

This bar chart compares total sales generated by different automobile brands. It clearly identifies top-performing brands and weaker performers in the market. The analysis helps management focus on successful brands and improve strategies for underperforming product.



Figure 4: Sales by Country

This geographical map visual displays the total sales contribution from different countries. It helps in understanding regional market demand, international business opportunities, and areas requiring better market penetration. It is highly useful for expansion and targeting strategies.

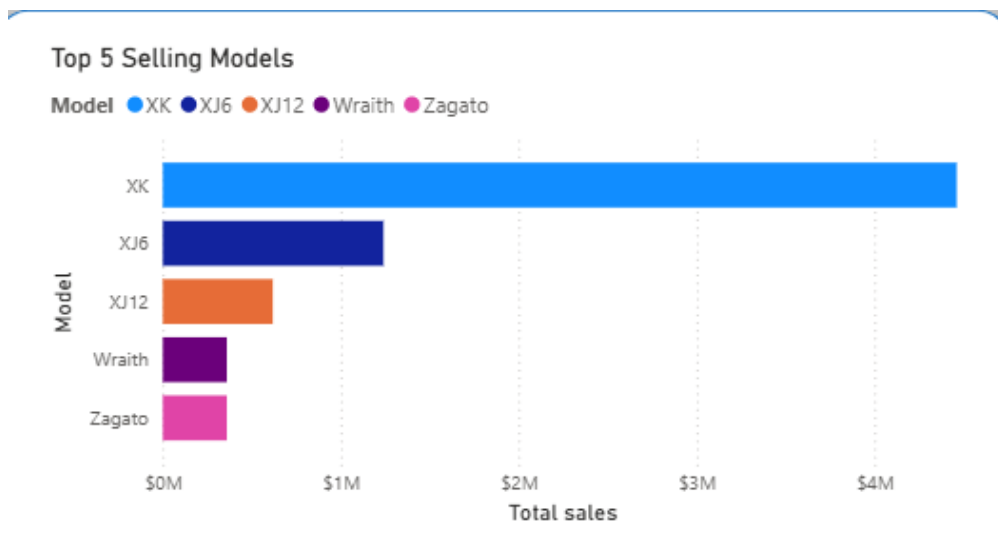


Figure 5: Top Models

This chart highlights the highest-selling vehicle models based on total revenue. It helps identify customer preferences and the most successful products in the portfolio. Business can use this information for marketing, stock planning, and future product development.

INSIGHTS

- Some automobile brands generate significantly higher sales than others.
- Certain countries contribute major revenue to total business sales.
- High discounts may reduce profitability.
- Monthly trends show variation in customer demand.
- A few models dominate total sales contribution.
- Profit margins differ across brands and markets.
- Interactive filters improve analysis by year, country, and brand.

CONCLUSION

The Automobile Dashboard Analysis project successfully demonstrates how raw business data can be transformed into meaningful insights using Microsoft Power BI. Through the use of data cleaning, DAX calculations, and interactive visualizations, the project provides a clear understanding of automobile sales performance across different brands, countries, customers, and time periods. It highlights the importance of business intelligence tools in modern decision-making processes.

The dashboard created in this project enables users to monitor key performance indicators such as Total Sales, Total Profit, Profit Margin, and Customer Count in a quick and efficient manner. Various charts and visuals help identify top-selling brands, profitable markets, seasonal sales trends, and customer preferences. This allows businesses to recognize strengths, weaknesses, and opportunities for improvement.

One of the major learnings from this project is that data alone has limited value unless it is properly analyzed and presented. By applying Power Query transformations and DAX formulas, the dataset became more useful and informative. Calculations such as running totals, growth percentages, moving averages, and rankings made the dashboard more dynamic and practical for business use.

The project also shows how discounts, delivery charges, spare parts costs, and labor costs affect final profitability. Such insights are valuable for management while planning pricing strategies,

controlling expenses, and improving operational efficiency. Geographic analysis through maps also helps identify high-performing countries and potential markets for expansion.

From a learning perspective, this project provided practical experience in Power BI dashboard development, data modeling, and report design. It improved analytical thinking, problem-solving ability, and understanding of how businesses use data to make strategic decisions.

In conclusion, the Automobile Dashboard Analysis project achieved its objective of converting raw automobile sales data into an organized and interactive reporting system. It proves that data visualization and analytics are powerful tools for improving business performance. The knowledge gained through this project can be applied to real-world scenarios in finance, sales, marketing, and operations management.